

Is KMAP Possible?

Formal Verification of Karnaugh Map Encoding for Biological Sequences in Lean 4

Shuvam Banerji Seal, Susmita Roy[†], Dwaipayan Roy

Indian Institute of Science Education and Research Kolkata, India

[†]Corresponding author: susmita.roy@iiserkol.ac.in

Companion to the poster

Every algorithm, definition, data set and number on the poster

What this document is. The poster states results in a few words. This companion gives, section by section in the poster's own order, the definitions behind every picture, the algorithms that produced every number, the data each number was counted on, and the exact mathematical statement of every one of the 131 machine-checked theorems (Appendix A). Every number here is a count taken exhaustively over a stated data set; none is an estimate, a sample or a fit. The figures are taken from the poster itself.

What it is not. The Lean 4 development (definitions and proofs) is private while the work is under review; it is not reproduced here. Each theorem is stated in ordinary mathematical notation instead, and every statement can be re-checked case by case in a web browser on the demo site.

Demo site (any PDB entry, from file to circuit, rules and inferences):

e-motioner-x-sbs.github.io/ICMS_Submission

Poster PDF and code:

github.com/E-Motioner-X-SBS/ICMS_Submission

* This work is currently under review.

Turn any protein
into a circuit



Any PDB entry, step by step,
to its rules and inferences.

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How to read the section numbers. The poster numbers its sections with a 3-bit Gray code: 000, 001, 011, 010, 110, 111, 101, 100. Each number differs from the one before in exactly one bit (the orange bit on the poster), which is the property the whole poster is about. This document follows that order and prints the same codes in its headings.

1 The poster at a glance



Abstract No.
ICMS2026-F-8
Theory and Methodology

The 7th International Conference on Molecular Simulation
4-8 October, 2026
Indian Institute of Science, Bengaluru



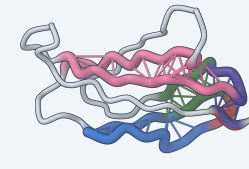
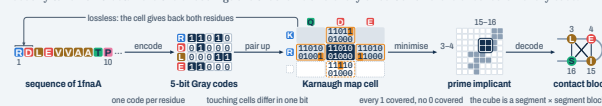
Is KMAP Possible? Formal Verification of Karnaugh Map Encoding for Biological Sequences in Lean 4

Shuvam Banerji Seal, Susmita Roy*, Dwaipayan Roy
Indian Institute of Science Education and Research Kolkata, India
*susmita.roy@iiserkol.ac.in

000 The finding*

We formalised in Lean 4 the mathematics that turns DNA, RNA and protein sequences into Karnaugh maps: 131 machine-checked theorems, none left unproved. On this foundation every structure becomes an exact Boolean circuit with exception-free rules. So does a viral species, given a dataset of its sequences: here 1,299 SARS-CoV-2 Omicron Spike proteins.

Gray-coded grids of sequences, from Petoukhov's genetic Karnaugh matrices¹ to chaos-game maps (FCGR)^{2,3}, are usually taken on trust. Here the encodings are and every circuit and rule is checked on every case.



Fihronectin type-III domain (PSICOV 1fnaA, 91 residues). Its exact contact circuit, read as rules: block rules that continue one another are strand pairs, one colour each.

Strand pairs, found as blocks
25-34 with 61-70 6 blocks
1-8 with 14-18 4 blocks
11-14 with 53-56 3 blocks
57-60 with 89-90 2 blocks
9-10 with 89-90 1 block

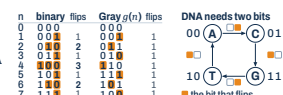
Turn any protein into a circuit



Any PDB entry, step by step, to its rules and inferences.

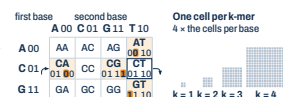
001 A code that changes one bit at a time

Binary counting can flip many bits at once: 0111 → 1000 flips four. The reflected Gray code $g(n) = n \oplus (n \gg 1)$ visits every number one and changes exactly one bit per step. DNA needs two bits, and A C G T is itself a Gray cycle.



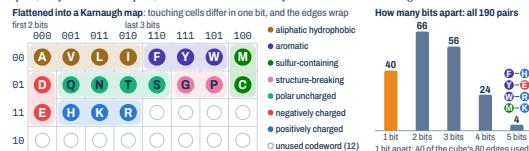
011 A Karnaugh map* is a folded truth table

Order a grid's rows and columns by the Gray code and touching cells differ in one variable, even across the edges, which wrap like a torus. Put a k -mer's first letters on the rows and the rest on the columns: each 2-mer owns one cell of a 4×4 map, each 4-mer one of a 16×16 map.



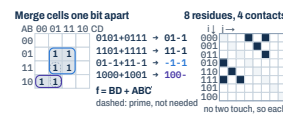
010 Amino acids on a five-dimensional cube

Proteins need five bits: 32 codewords for 20 residues. Numbering residues group by group before Gray-coding keeps chemistry close. All $20 \times 19 / 2 = 190$ pairwise distances are proved: 40 pairs sit one bit apart, only four sit five apart, and the code uses exactly half of the cube's 80 edges.



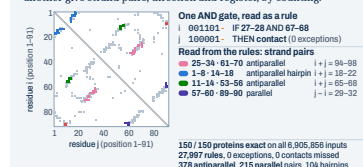
110 Contacts as a Boolean function

Contact map (Boolean function): $C(i,j) = 1 \Leftrightarrow \text{CP-CP} < 8 \text{ \AA}, |i-j| \geq 6$.
• **Quine-McCluskey**⁴: merge cells one bit apart.
• **Prime implicant**: a cube that cannot grow.
• **Circuit**: OR of covering primes = the map.
• **Lean**, 8 residues: symmetric, no merges.

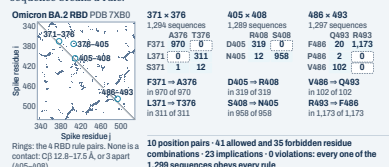


100 Rules with no exception, from exact circuits

Each PSICOV⁵ protein (150 Pfam families) becomes a circuit equal to its contact map. Its AND gates are rules; block rules that continue one another give strand pairs, direction and register, by counting.



One species, SARS-CoV-2 Omicron: in 1,299 Spike sequences, exact circuits list which residue pairs occur at co-evolving positions and which never do. No sequence breaks a rule.



Takeaways

- A verified foundation. The encodings are Lean theorems; every circuit is exact on all inputs.
- Rules that are true, not likely. Each holds for every contact, sequence or frame it names.
- One language for sequence, structure and dynamics. Any finite encoding can be checked the same way.
- Lossless by proof. Each cell decodes to both residues; Gray order only permutes cells, so no statistic changes.
- Exact about absence. No rule on chemistry alone holds without exceptions: all 196 group × separation chains of PSICOV pairs mix contacts and non-contacts.

Demo, any PDB entry: e-motifone-e-ss.github.io/ICMS_Submission

1. S. V. Petoukhov, *Bioinformatics* 2024, 246, 105349.
2. B. J. Jeffrey, *Nucleic Acids Res.* 1990, 18, 2163-2170.
3. J. S. Altmann, S. Vinga, *BMC Bioinformatics* 2009, 10, 100.

4. M. Karnaugh, *Trans. AIEE* 1953, 72, 593-599.
5. W. Y. Quine, *Am. Math. Mon.* 1952, 59, 521-531; E. J. McCluskey, *Bell Syst. Tech. J.* 1956, 35, 1417-1444.

6. L. de Moura, S. Ullrich, *CADE* 20, LNCS 12499, 2021, 425-435.
7. D. T. Jones et al., *Bioinformatics* 2012, 28, 184-190.
8. J. K. Noel et al., *Plant Comput. Biol.* 2014, 12, e004794.

* This work is currently under review.

The A0 poster (841 × 1189 mm). Reading order: band at the top (000), left column top to bottom (001, 011, 010, 110), right column top to bottom (111, 101 and the folding-simulation panel), band at the bottom (100), references.

2 Notation and conventions used throughout

- $[n] = \{0, 1, \dots, n-1\}$. Positions inside a chain are counted from 0 in the bits and in all formulas; residue labels in the text and on the poster are counted from 1, as in the PDB file, unless stated.
- $\oplus$ is bitwise exclusive or; $d(a, b)$, the Hamming distance, is the number of bits in which a and b differ.
- The reflected binary Gray code is $g(n) = n \oplus \lfloor n/2 \rfloor$; its inverse is $\bar{g}(m) = m \oplus \lfloor m/2 \rfloor \oplus \lfloor m/4 \rfloor \oplus \dots$.
- DNA letters get the 2-bit codes $e(A) = 00$, $e(C) = 01$, $e(G) = 11$, $e(T) = 10$ (RNA: U takes T's code). Amino acids are numbered 0–19 in the group order of table 2 and get the 5-bit code $\gamma(a) = g(e(a))$.
- A *cube* (an *implicant*) over n bits is a pair (v, m) of a value and a mask: it contains every x with $x \wedge \neg m = v$, i.e. x agrees with v on every bit not set in m . The bits set in m are *free*; a cube with u free bits contains 2^u points and is written as a pattern of 0, 1 and - (free). In circuit terms a cube is one AND gate.
- *Exact* means: the circuit was evaluated on every one of its inputs and agrees with the data everywhere. *Sound*: no gate covers a 0. *Complete*: every 1 is covered.

3 000 The finding

Poster text. We formalised in Lean 4 the mathematics that turns DNA, RNA and protein sequences into Karnaugh maps: 131 machine-checked theorems, none left unproved. On this foundation every structure becomes an exact Boolean circuit with exception-free rules. So does a viral species, given a dataset of its sequences: here 1,299 SARS-CoV-2 Omicron Spike proteins.

Gray-coded grids of sequences have a long history, from Petoukhov's genetic Karnaugh matrices [1] to the chaos-game representation of DNA (FCGR) [2, 3]. The properties these grids rely on (that neighbouring cells differ in one letter, that no information is lost, that a block of cells is a stretch of sequence) are usually taken on trust. Here each of them is a theorem, and every circuit and rule built on them is checked on every case.

3.1 The flowchart: one protein through the whole pipeline

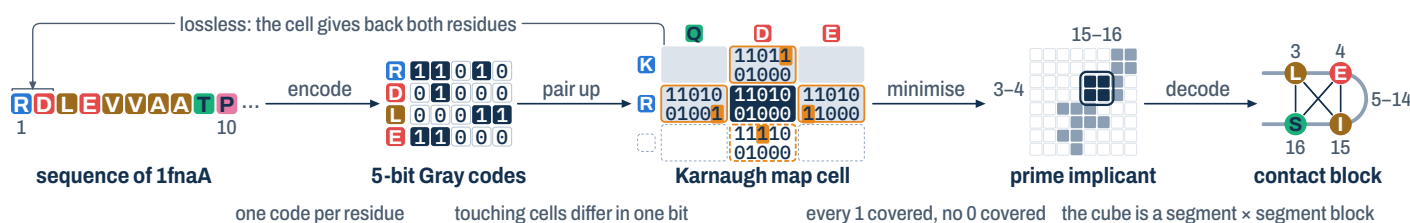


Figure 1. The flowchart of band 000. Gloss under each arrow: what the corresponding theorem guarantees.

The flowchart follows the first residues of PSICOV target 1fnaA (the tenth fibronectin type-III domain, 91 residues), the protein drawn in 3D in the top-right corner of the poster.

1. **Sequence.** R D L E V V A A T P ..., one tile per residue, coloured by chemical group (table 2).
2. **Encode.** Each residue gets its 5-bit code γ : $R \mapsto 11010$, $D \mapsto 01000$, $L \mapsto 00011$, $E \mapsto 11000$. *Guarantee*: one code per residue (the code is injective, T52).
3. **Pair up (Karnaugh map cell).** A pair of residues is a cell of the 32×32 residue-pair Karnaugh map: row = code of the first residue, column = code of the second, both axes in Gray order. The dipeptide

RD is the cell (row 11010, column 01000); because $\gamma(a) = g(e(a))$, residue number k sits at axis position k , so RD is at position (19, 15). The magnified 3×3 window shows its four touching cells, each with its own two codes: row K (11011) above, the unused codeword 11110 below, column Q (01001) to the left, column E (11000) to the right. The orange bit is the one that differs from RD's codes: *touching cells differ in one bit*. Dashed cells are unused codewords (padding, §9).

4. **Minimise (prime implicant).** An 8×8 window (residues 1–8 against 11–18) of the real 1fnaA contact map. Contacts are grey; the outlined cube is one prime implicant of the minimised contact function, the block of residues 3–4 against 15–16. *Guarantee:* the cover covers every 1 and no 0.
5. **Decode (contact block).** The cube, read back, is two chain segments that touch: residues 3–4 and 15–16, with all four contacts (L3–I15, L3–S16, E4–I15, E4–S16 in the drawing). *Guarantee:* a cube with low free bits in both position fields is a segment $\times$ segment block (T121).
6. **Return path.** The cell gives back both residues: packing (position, residue) into one number is injective (T124), and so is the code (T52). The arrow is drawn from the Karnaugh-map cell, not from the contact block, because a contact map alone does not determine the sequence.

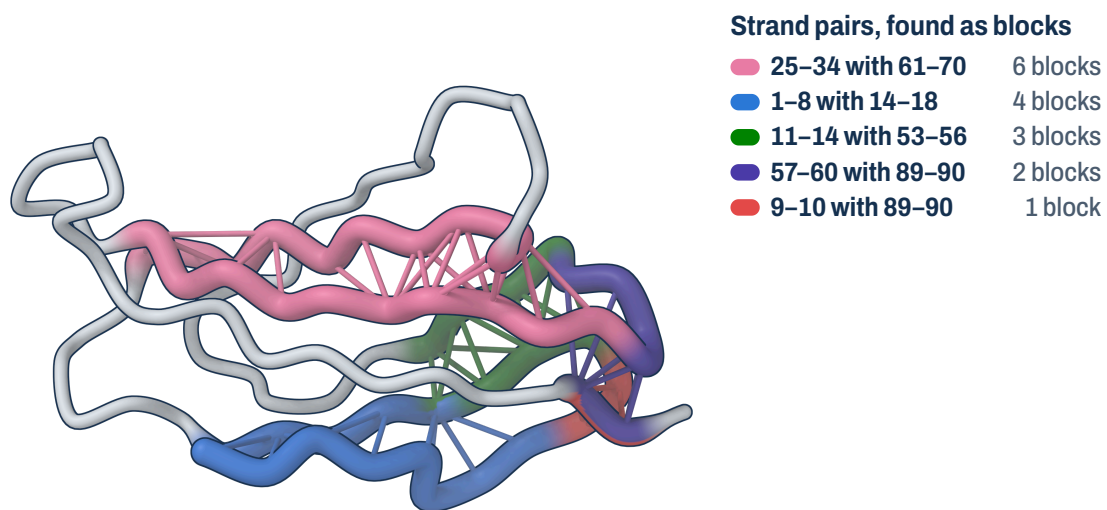


Figure 2. 1fnaA in 3D (from the poster). Block rules that continue one another are strand pairs, one colour each; the lines inside each ribbon pair are the contacts named by the rules.

Say it at the poster

“Every step from a sequence to a circuit is a theorem: the codes are distinct, neighbouring cells differ in one bit, the circuit is checked on every input, and a block of cells really is two pieces of chain that touch.”

4 001 A code that changes one bit at a time

Counting in binary can flip many bits at once ($0111 \rightarrow 1000$ flips four). The reflected Gray code $g(n) = n \oplus \lfloor n/2 \rfloor$ visits every number once and changes exactly one bit per step.

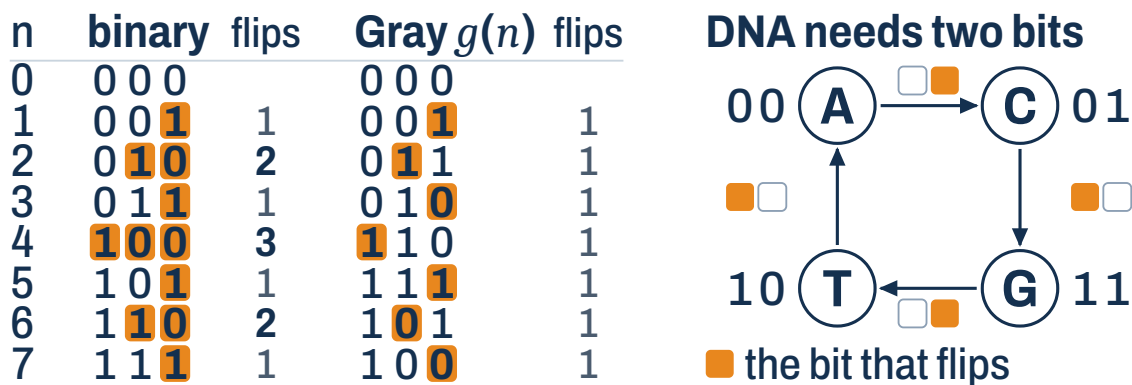


Figure 3. Section 001 of the poster: binary and Gray counting from 0 to 7 with the number of flipped bits per step, and the four DNA letters on the 2-bit Gray cycle $A \rightarrow C \rightarrow G \rightarrow T \rightarrow A$.

Algorithm 1 Gray code and its inverse

```

1: function GRAY( $n$ )
2:   return  $n \oplus (n \gg 1)$ 
3: function GRAYINVERSE( $m$ )
4:    $v \leftarrow m; s \leftarrow 1$ 
5:   while  $s < \text{word length}$  do
6:      $v \leftarrow v \oplus (v \gg s); s \leftarrow 2s$ 
7:   return  $v$ 

```

▷ prefix XOR: bit k of the result is the XOR of bits $\geq k$ of m

Table 1. The first sixteen numbers: binary and Gray, with the number of bits that change from the previous row.

n	binary	bits flipped	Gray $g(n)$	bits flipped
0	0000		0000	
1	0001	1	0001	1
2	0010	2	0011	1
3	0011	1	0010	1
4	0100	3	0110	1
5	0101	1	0111	1
6	0110	2	0101	1
7	0111	1	0100	1
8	1000	4	1100	1
9	1001	1	1101	1
10	1010	2	1111	1
11	1011	1	1110	1
12	1100	3	1010	1
13	1101	1	1011	1
14	1110	2	1001	1
15	1111	1	1000	1

What is proved (Appendix A, “Gray code and the DNA letters” and “k-mer Karnaugh maps”). For every $n < 255$, $g(n)$ and $g(n+1)$ differ in exactly one bit (T2, T3); g is injective on 8 bits (T4) and $\bar{g}(g(n)) = n$ (T5). On two bits the cycle closes: $g(3)$ and $g(0)$ also differ in one bit (T20), so the edges of a Karnaugh map wrap around. **DNA needs two bits**, and with the codes $A = 00$, $C = 01$, $G = 11$, $T = 10$ the order A, C, G, T is itself a Gray cycle (T9–T12); the four codes are distinct (T6, T16).

Say it at the poster

“Gray order is what makes a Karnaugh map work: neighbours differ in one variable, even across the edges. For DNA the alphabetical-looking order A, C, G, T already is a Gray cycle once G is 11 and T is 10.”

5 011 A Karnaugh map[4] is a folded truth table

A Karnaugh map lays the 2^n rows of a truth table on a grid whose rows carry the high bits and whose columns carry the low bits, each axis in Gray order. Touching cells then differ in exactly one variable, even across the edges, which wrap like a torus. For sequences, a k -mer's first letters go on the rows and the rest on the columns: each 2-mer owns one cell of a 4×4 map, each 4-mer one cell of a 16×16 map; in general a k -mer of DNA has $2k$ bits and a $2^k \times 2^k$ map.

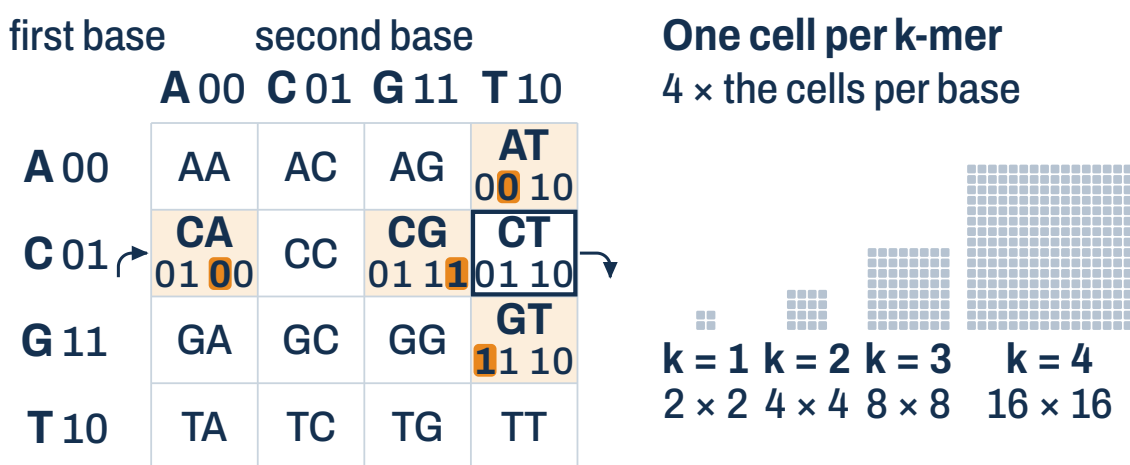


Figure 4. Section 011: the 4×4 dinucleotide map with Gray-ordered axes (A, C, G, T $\mapsto$ 00, 01, 11, 10), cells one bit apart highlighted, and the map sizes for $k = 1$ to 4.

Algorithm 2 The Karnaugh-map cell of a k -mer

Input: letters $x_1 \dots x_k$, letter code e of b bits ($b = 2$ for DNA/RNA, 5 for proteins)

- 1: $c \leftarrow 0$; **for** $t = 1..k$: $c \leftarrow c \cdot 2^b + e(x_t)$ ▷ first letter most significant: E_k
- 2: $n \leftarrow kb$; $r_b \leftarrow \lfloor n/2 \rfloor$; $c_b \leftarrow n - r_b$ ▷ row bits, column bits
- 3: row label $\leftarrow c \gg c_b$; column label $\leftarrow c \wedge (2^{c_b} - 1)$
- 4: **return** display position $(x, y) = (\text{GRAYINVERSE}(\text{column}), \text{GRAYINVERSE}(\text{row}))$

Counting overlapping k -mers of a sequence into these cells gives the sequence's Karnaugh map; windows that cross a strand junction or a gap in the modelled chain are skipped. **Proved:** the 16 dinucleotides, 64 trinucleotides and 256 tetranucleotides land in distinct cells inside the map (T38–T48); row and column together determine a dinucleotide (T44); the cell is $4 \times \text{row} + \text{column}$ (T37); on the 2×2 map A–C and G–T sit side by side and C–G and T–A one above the other, so the cycle walks round the square (T33–T36).

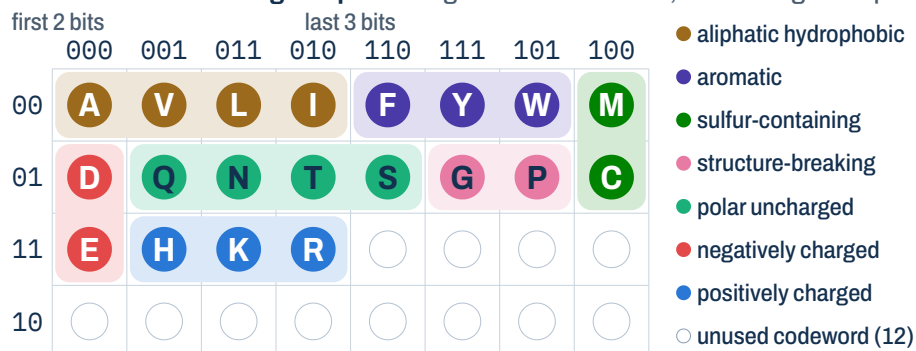
6 010 Amino acids on a five-dimensional cube

Proteins need five bits: 32 codewords for 20 residues. Numbering the residues group by group before Gray coding keeps chemistry close: the code of residue number k is $g(k)$, so consecutive residues in the order of table 2 are always one bit apart (T106).

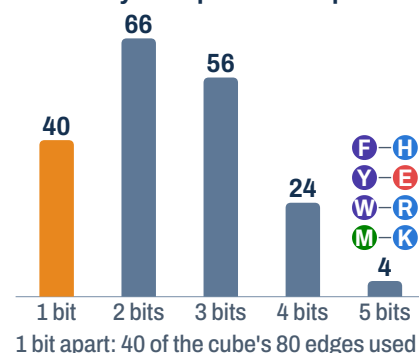
Table 2. The amino-acid code: group order, raw index $e(a)$ and 5-bit code $\gamma(a) = g(e(a))$. The 12 codewords 16–23 and 28–31 are unused.

group	residue	$e(a)$	binary $e(a)$	code $\gamma(a)$
aliphatic hydrophobic	A	0	00000	00000
	V	1	00001	00001
	L	2	00010	00011
	I	3	00011	00010
aromatic	F	4	00100	00110
	Y	5	00101	00111
	W	6	00110	00101
sulfur-containing	M	7	00111	00100
	C	8	01000	01100
structure-breaking	P	9	01001	01101
	G	10	01010	01111
polar uncharged	S	11	01011	01110
	T	12	01100	01010
	N	13	01101	01011
	Q	14	01110	01001
negatively charged	D	15	01111	01000
	E	16	10000	11000
positively charged	H	17	10001	11001
	K	18	10010	11011
	R	19	10011	11010

Flattened into a Karnaugh map: touching cells differ in one bit, and the edges wrap



How many bits apart: all 190 pairs

**Figure 5.** Section 010: the 20 codes flattened into a 4×8 Karnaugh map (first 2 bits on rows, last 3 bits on columns, both Gray-ordered), and the distribution of code distances over all 190 pairs.

Every pairwise distance is proved. There are $20 \times 19/2 = 190$ unordered pairs of amino acids. Their code distances are: 40 pairs at distance 1, 66 at 2, 56 at 3, 24 at 4 and only 4 at 5 (T101); the four maximally distant pairs are F–H, Y–E, W–R and M–K (T68–T71). Over ordered pairs the counts double (80, 132, 112, 48, 8, T100), and with the 20 pairs of a residue with itself the ordered distribution over all 400 pairs is (20, 80, 132, 112, 48, 8) for $d = 0, \dots, 5$ (T110). The five-dimensional cube has $5 \cdot 32/2 = 80$ edges (T104); the code uses 40 of them, exactly half (T105), within the bound of 50 that any 20 corners of the cube allow (T103). Within groups: A, V, L, I lie within two bits of one another, as do F, Y, W and H, K, R; M–C, P–G and D–E are one bit apart; S, T, N, Q form a path of one-bit steps whose ends are three bits apart (T54–T66, T74). Crossing groups is possible too: there are 26 one-bit pairs across groups, all listed (T75–T78).

Say it at the poster

“Twenty residues need five bits. Ordering them by chemistry before Gray coding puts chemical neighbours on neighbouring corners of the 5-cube. All 190 distances are proved, not sampled.”

7 110 Contacts as a Boolean function

Contact map (Boolean function). For a protein chain of length L , $C(i, j) = 1$ exactly when the representative atoms of residues i and j are closer than $8 \text{ \AA}$ and $|i - j| \geq 6$: $C\beta$ atoms, $C\alpha$ for glycine (the PSICOV convention [7]). With $p = \lceil \log_2 L \rceil$ bits per position, C is a Boolean function of $2p$ variables: the high p bits are i , the low p bits are j , positions in plain binary. Inputs with $i \geq L$ or $j \geq L$ (padding) are 0.

Algorithm 3 Contact map and contact function of a chain

Input: residues with coordinates; cutoff r_c ($8 \text{ \AA}$ protein; $10 \text{ \AA}$ DNA/RNA on $C1'$, P if missing); minimum separation s (6 protein; 3 DNA/RNA)

- 1: **for** each pair $i < j$ with a representative atom **do**
- 2: **if** i, j on the same strand **and** their sequence numbers differ by $< s$ **then continue**
- 3: **if** $\|x_i - x_j\| < r_c$ **then** add contact (i, j)
- 4: $p \leftarrow \max(1, \lceil \log_2 \max(L, 2) \rceil)$
- 5: on-set $\leftarrow \{(i \ll p) \vee j, (j \ll p) \vee i \mid (i, j) \text{ a contact}\}$ $\triangleright$ both orientations: the map is symmetric
- 6: **return** the function of $2p$ variables that is 1 on the on-set and 0 everywhere else

Key terms.

- *Merge* (Quine–McCluskey [5]): two cubes with the same free bits whose values differ in one bit merge into a cube with one more free bit; repeated until nothing merges.
- *Prime implicant*: a cube that lies inside the on-set and cannot grow.
- *Cover / circuit*: a set of primes covering every 1. Their OR is a two-level AND–OR circuit equal to the map.

Algorithm 4 All prime implicants (exact Quine–McCluskey)

Input: on-set M of an n -variable function (no don't-cares in this work)

- 1: $\mathcal{C} \leftarrow \{(m, 0) : m \in M\}$; primes $\leftarrow \emptyset$
- 2: **while** $\mathcal{C} \neq \emptyset$ **do**
- 3: bucket the cubes of $\mathcal{C}$ by (mask, number of 1s in value)
- 4: **for** each cube (v, m) and each free-able bit b with $b \notin m, b \notin v$ **do**
- 5: **if** $(v \vee b, m) \in \mathcal{C}$ **then** $\triangleright$ a partner in the next bucket, one bit apart
- 6: add $(v, m \vee b)$ to $\mathcal{C}'$; mark (v, m) and $(v \vee b, m)$ as used
- 7: primes $\leftarrow$ primes $\cup$ {unused cubes of $\mathcal{C}$ }; $\mathcal{C} \leftarrow \mathcal{C}'$
- 8: **return** primes, sorted by (value, mask)

Algorithm 5 The cover: essential primes, then greedy completion

- 1: for each 1-cell, count the primes that contain it
- 2: $\mathcal{E} \leftarrow$ primes that are the only cover of some 1-cell ▷ essential: in every cover
- 3: $S \leftarrow \mathcal{E}$; $U \leftarrow$ 1-cells not covered by $\mathcal{E}$
- 4: **while** $U \neq \emptyset$ **do**
- 5: pick the prime covering the most cells of U (first in sorted order on ties); add it to S ; remove its cells from U
- 6: check: every 1-cell in some cube of S (complete) and every point of every cube is a 1 (sound)
- 7: **return** S ▷ exact; minimum only when $S = \mathcal{E}$

Algorithm 6 Run the circuit on every input

- 1: $O[0..2^{2p} - 1] \leftarrow 0$
- 2: **for** each AND gate (v, m) of the cover **do**
- 3: $t \leftarrow m$; **repeat** $O[v \vee t] \leftarrow 1$; **if** $t = 0$ **break**; $t \leftarrow (t - 1) \wedge m$ ▷ every point of the cube
- 4: **for** every input $x = (i \ll p) \vee j$ **do** compare $O[x]$ with $C(i, j)$ (0 for padding)
- 5: **return** exact $\iff$ all 2^{2p} inputs agree

An honest caveat. The cover is exact (sound and complete, verified on every input) but it is *not certified minimal*: the greedy completion can use more gates than necessary. Minimality is claimed only where the cover consists of essential primes alone (the Spike circuits, §11.5).

Merge cells one bit apart

AB	00	01	11	10	CD
00					
01		1	1		
11		1	1		
10	1	1			

$$0101 + 0111 \rightarrow 01-1$$

$$1101 + 1111 \rightarrow 11-1$$

$$01-1 + 11-1 \rightarrow -1-1$$

$$1000 + 1001 \rightarrow 100-$$

$$f = BD + A\bar{B}\bar{C}$$

dashed: prime, not needed

8 residues, 4 contacts

i ↓ j →	000	001	011	010	110	111	101	100
000								
001								
011								
010								
110								
111								
101								
100								

no two touch, so each

is its own prime implicant

Figure 6. Section 110. Left: Quine–McCluskey on $f(A, B, C, D) = \sum m(5, 7, 8, 9, 13, 15)$. The merges $0101 + 0111 \rightarrow 01-1$, $1101 + 1111 \rightarrow 11-1$, $01-1 + 11-1 \rightarrow -1-1$ (BD) and $1000 + 1001 \rightarrow 100-$ ($A\bar{B}\bar{C}$) give $f = BD + A\bar{B}\bar{C}$; the prime $1-01$ ($A\bar{C}D$, dashed) is not needed. Right: the 8-residue example map of the Lean development, contacts $(0,3)$, $(1,4)$, $(2,5)$, $(0,5)$ and mirrors, on Gray-indexed rows.

The 8-residue example. Lean proves the example map symmetric and irreflexive (T79, T80), lists its contacts (T81, T92) and classifies all 64 entries (T93). With Gray-indexed cells $c_{ij} = 8g(i) + g(j)$ the four contacts sit at cells 2, 14, 31 and 7 (T90), pairwise 2, 4, 2, 2, 2, 2 bits apart (T97): no two contact cells are one bit apart (T96), so Quine–McCluskey cannot merge any of them and every contact is its own prime implicant.

8 111 How Lean checks a claim

- **Lean 4** [6]: a statement is a type, a proof is a program of that type, and a small kernel checks that the program has the type.
- **Compiled exhaustive check** (Lean’s `native_decide`): for a statement over finitely many cases the proof runs every case in compiled code, at most 2^{20} cases per theorem here. Such a proof also trusts Lean’s compiler.
- **0 unproved steps** (no sorry) and **0 axioms of our own**.

The poster prints one theorem as an example: Gray coding is additive over GF(2),

$$\forall x, y \in [32] : g(x \oplus y) = g(x) \oplus g(y) \quad (\text{T127, proved by evaluating all 1,024 cases}).$$

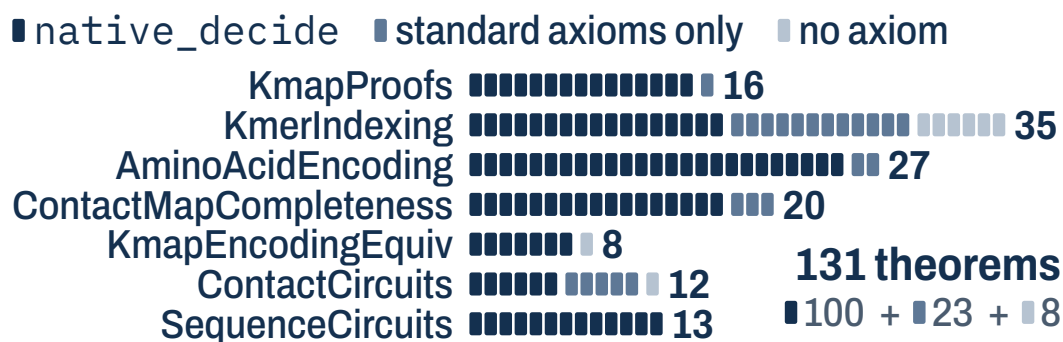


Figure 7. The inventory of the 131 theorems by what the proof trusts, as printed on the poster. The bars are labelled by source file; in this document the files are the topics of Appendix A, in the same order: Gray code and the DNA letters, *k*-mer Karnaugh maps, the amino-acid code, an 8-residue contact map, the five-bit cube, contact circuits, sequence circuits.

Table 3. The 131 theorems by what their proofs trust.

proof	theorems	trusts
compiled check of every case (directly or through a lemma)	100	Lean’s kernel, its standard axioms and its compiler
kernel-checked tactics	23	Lean’s kernel and its standard axioms
kernel-checked, no axioms at all	8	Lean’s kernel only

By topic: Gray code and the DNA letters 16; *k*-mer Karnaugh maps 35; the amino-acid code 27; the 8-residue contact map 20; the five-bit cube 8; contact circuits 12; sequence circuits 13. All statements are in Appendix A. The demo site re-runs the finite statement of every theorem in the browser: 9,323,294 cases in all (fewer when a search for a witness stops early). A browser check is a re-computation of the statement, not a proof; the proofs are Lean’s.

9 101 What the proofs guarantee

9.1 Exact rules on padded maps

Twenty residues need five bits per axis, so a 20×20 residue-pair map is padded to 32×32 , with 10-bit addresses $32v + r$ (row v , column r). A rule with all ten bits fixed is the cube $Q_{10}(32v + r, 0)$. Lean proves that such a rule matches exactly its own cell, for all $v, r < 32$ and every address (T117, 1,048,576 cases), and that a fully fixed rule about two real residues ($v, r < 20$) fires only inside the 20×20 block, never on padding (T118).

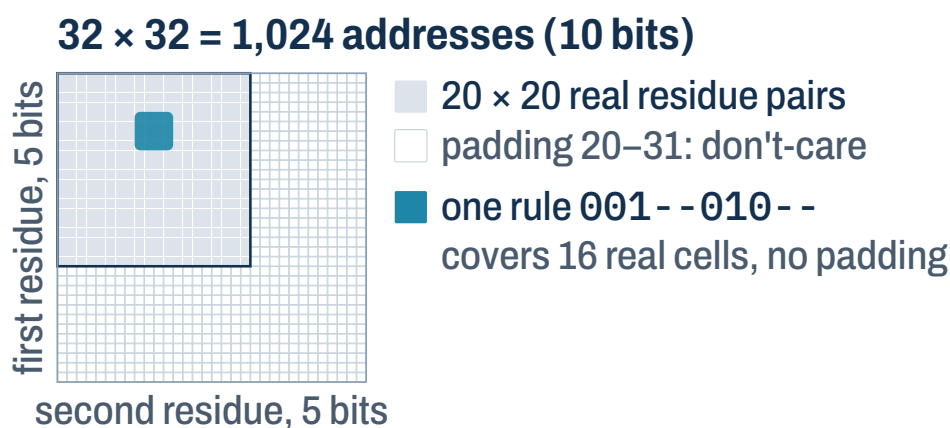


Figure 8. The padded 32×32 grid: real 20×20 residue pairs, padding rows and columns 20–31, and one rule 001 - - 010 - - that covers 16 real cells and no padding.

9.2 Segments hold under either numbering

With plain-binary positions, a cube whose free bits are the low u bits and whose value v is a multiple of 2^u is exactly the unbroken stretch $v, v + 1, \dots, v + 2^u - 1$ of the chain (the segment lemma, T119; it covers exactly 2^u positions, T120). A contact-map cube with low free bits in both fields is therefore a block, one segment against another (T121). Under Gray-coded positions the same cube is still an unbroken stretch (T122), but not always the same one (T123): binary and Gray name different segments, never broken ones, so the two must never be mixed.

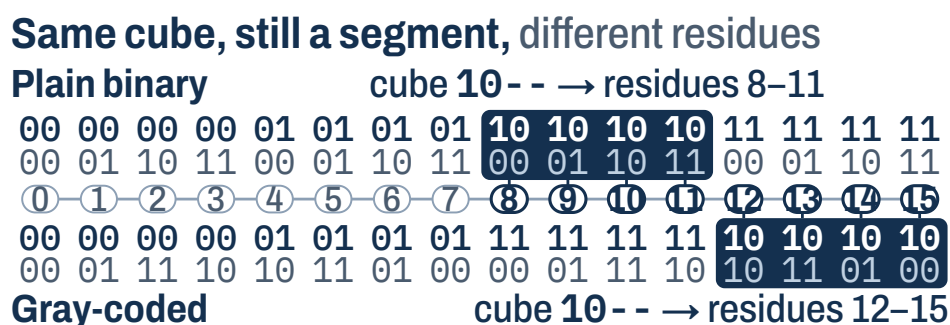


Figure 9. The same cube 10 - - names residues 8–11 under plain binary and 12–15 under Gray-coded positions.

9.3 What the encoding can and cannot change

Since $g(x \oplus y) = g(x) \oplus g(y)$ (T127, T128) and g is a bijection on five bits (T131), re-indexing the rows and columns of a map by g only permutes its cells. A Gray-ordered map is therefore a lexicographic (FCGR) map with its cells moved: every statistic of the cell values is identical. Measured on 120 protein maps, the cophenetic correlation of hierarchical clustering is 0.976747 under both orderings, and the sorted cell values are identical. The code changes only the distances between letters: of the 400 ordered amino-acid pairs, exactly 80 are one bit apart (T109), and some pairs of numbers sit at a different Hamming distance after Gray coding (T130).

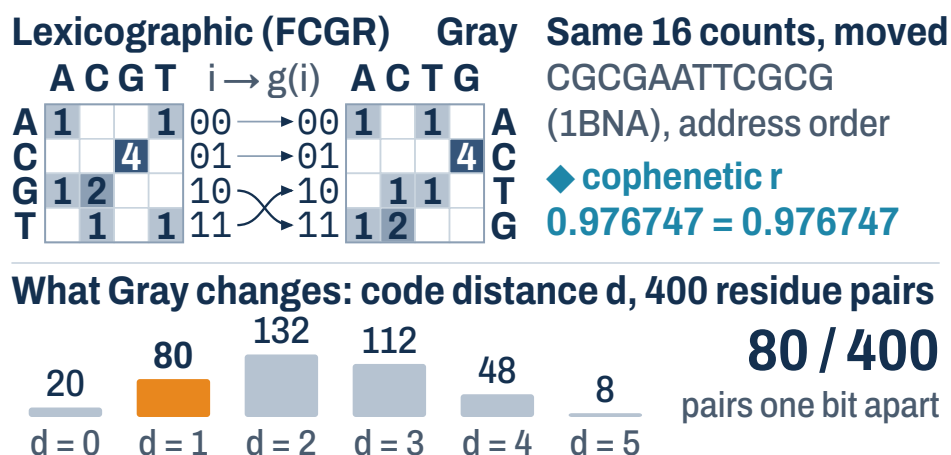


Figure 10. Top: the dinucleotide counts of CGCGAATTCGCG (PDB 1BNA, one strand) on a lexicographic and a Gray 4×4 map: the same 16 counts, moved. Bottom: the 400 ordered amino-acid pairs by code distance $d = 0$ to 5.

Say it at the poster

“The Gray code does not add or hide information: it permutes cells. What it changes is which cells touch, and that is exactly what lets Quine–McCluskey merge neighbours into segments.”

10 Rules that hold in every frame of a folding run

Data. Structure-based-model (SMOG2, $C\alpha$ model [8]) folding trajectories of 479 runs (436 unique proteins), 48,385 frames in all. A native contact is *formed* in a frame when its $C\alpha$ distance is below $1.2 \times$ its native distance, otherwise *broken*. Each trajectory is therefore a binary frames $\times$ contacts matrix S .

Rule. “Whenever contact a is formed, so is contact t ” ($a \Rightarrow t$) is kept when it holds in *every* frame (zero counterexamples) and is not trivial: a is formed in at least 20 frames and t is broken in at least 5 frames (trajectories of at least 50 frames). Two contacts (i, j) and (k, l) are *neighbours* on the contact map when $|i - k| + |j - l| \leq 2$.

Algorithm 7 Every-frame implications of a trajectory

Input: $S \in \{0, 1\}^{F \times K}$ (frames $\times$ contacts that vary)

- 1: $n_1 \leftarrow$ column sums of S (frames in which each contact is formed); $co \leftarrow S^T S$
- 2: **for** each ordered pair (a, t) , $a \neq t$ **do**
- 3: **if** $co[a, t] = n_1[a]$ **and** $n_1[a] \geq 20$ **and** $F - n_1[t] \geq 5$ **then**
- 4: keep $a \Rightarrow t$ ▷ every frame with a formed has t formed
- 5: two-input rules $a \wedge b \Rightarrow t$: kept only if they hold in every frame and neither $a \Rightarrow t$ nor $b \Rightarrow t$ does
- 6: re-count each kept rule frame by frame: counterexamples must be 0

Table 4. Every-frame rules. Columns: runs, frames, runs with at least one rule, implications $a \Rightarrow t$, implications joining neighbouring contacts, equivalences ($a \Leftrightarrow t$), two-input rules.

set	runs	frames	with rules	$a \Rightarrow t$	neighbours	$\Leftrightarrow$	$a \wedge b \Rightarrow t$
all runs	479	48,385	474	9,986	3,966	7	0
unique proteins	436	44,085	433	8,875	3,661	6	0
dense 8,000-frame runs	10	80,000	8	33	33	0	0

In the ten long runs of 8,000 frames, every one of the 33 every-frame rules joins neighbouring contacts.

Crambin (1CBN), the poster’s example: 103 native contacts, 97 of which vary; the single rule is contact (4, 44) $\Rightarrow$ contact (4, 43). Its frame table over 8,000 frames:

	t formed	t broken
a formed	5,429	0
a broken	1,258	1,313

The zero is the rule: in no frame is a formed while t is broken.

Table 5. One rule from each dense run that has rules, with its frame table ($a \& t$ / $a \& \neg t$ / $\neg a \& t$ / $\neg a \& \neg t$).

run	frames	rules	neighbour rules	example $a \Rightarrow t$	frame table
1m1q	8,000	7	7	(56,60) $\Rightarrow$ (55,60)	4,995 / 0 / 1,515 / 1,490
1j0o	8,000	5	5	(67,71) $\Rightarrow$ (67,72)	6,398 / 0 / 967 / 635
1m1r	8,000	7	7	(56,60) $\Rightarrow$ (55,60)	4,931 / 0 / 1,656 / 1,413
1j0p	8,000	5	5	(68,72) $\Rightarrow$ (68,73)	6,804 / 0 / 766 / 430
1jxy	8,000	2	2	(9,30) $\Rightarrow$ (8,30)	6,080 / 0 / 1,033 / 887
1jxu	8,000	1	1	(4,44) $\Rightarrow$ (4,43)	5,207 / 0 / 1,228 / 1,565
1CBN	8,000	1	1	(4,44) $\Rightarrow$ (4,43)	5,429 / 0 / 1,258 / 1,313
1C75	8,000	5	5	(39,64) $\Rightarrow$ (39,63)	6,289 / 0 / 1,310 / 401

Crambin, 8,000 frames

a = contact 4–44, t = 4–43

	t formed	t broken
a formed	5,429	0
a broken	1,258	1,313

9,986

rules hold in every frame

of 479 folding runs

like $a \Rightarrow t$: a formed
never without t

3,966 join neighbouring contacts

33 / 33 in the long runs

Figure 11. The folding-simulation panel of the poster.

11 100 Rules with no exception, from exact circuits

11.1 From circuit to rules

Every AND gate of an exact contact circuit, read in words, is a rule: “IF $i \in I$ AND $j \in J$ THEN residues i and j touch”, where I and J are the positions admitted by the gate’s two fields. Because the cover is sound, no rule has an exception; because it is complete, together the rules name every contact.

Algorithm 8 Rules from the AND gates

- 1: **for** each gate (v, m) of the cover **do**
 - 2: $I \leftarrow \{x < L : x \text{ matches the high field}\}, J \leftarrow \{x < L : x \text{ matches the low field}\}$
 - 3: orient so that $\min I \leq \min J$; merge with an earlier rule having the same two patterns (its mirror image)
 - 4: cells $\leftarrow \{\{i, j\} : i \in I, j \in J, i \neq j\}$; exceptions $\leftarrow$ cells that are not contacts
 - 5: kind $\leftarrow$ **strided** if a field’s free bits are not the low ones; else **block** (≥ 2 free bits), **pair** (1), **single** (0)
 - 6: check: exceptions = 0 for every rule, and every contact is in some rule’s cells
-

11.2 From rules to inferences

Block rules that continue one another describe one pair of touching strands, and counting their cells gives the direction and register of the pairing.

Algorithm 9 Strand pairs, direction, register and hairpins

- 1: blocks $\leftarrow$ the block rules as rectangles $[i_0, i_1) \times [j_0, j_1)$ above the diagonal
 - 2: join two blocks (union-find) when *both* their i -ranges and their j -ranges overlap, touch or are at most 1 residue apart
 - 3: **for** each cluster with contacts P **do**
 - 4: $\Delta_+ \leftarrow \max_P(i + j) - \min_P(i + j)$; $\Delta_- \leftarrow \max_P(j - i) - \min_P(j - i)$
 - 5: **if** one block **then** direction $\leftarrow$ *short* (a rectangle has no direction)
 - 6: **else if** $\Delta_+ < \Delta_-$ **then** direction $\leftarrow$ *antiparallel*; register $\leftarrow$ range of $i + j$
 - 7: **else if** $\Delta_- < \Delta_+$ **then** direction $\leftarrow$ *parallel*; register $\leftarrow$ range of $j - i$
 - 8: **else** direction $\leftarrow$ *mixed*
 - 9: hairpin $\leftarrow$ antiparallel, same strand, and at most 8 residues between the two segments (by sequence number)
-

11.3 150 proteins (PSICOV, 150 Pfam families)

Data. The PSICOV benchmark [7]: 150 single-domain proteins of known structure, each from a different Pfam family; native contacts as in §7. Total 21,837 residues and 47,430 contacts.

Table 6. Exact circuits, rules and inferences over the 150 PSICOV proteins (every number a count).

circuits exact on every input	150 of 150
inputs evaluated ($\sum 2^{2^p}$)	6,905,856
1-cells (contacts in both orientations)	94,860
prime implicants / gates in the covers / of which essential	68,544 / 54,229 / 45,822
literals (wires into the AND gates)	792,362
rules (mirror images merged): block / pair / strided / single	27,997: 2,476 / 6,705 / 12,015 / 6,801
rules with an exception / contacts no rule names	0 / 0
contacts inside block rules / proteins with at least one block	9,260 / 145
strand pairs: antiparallel / parallel / short / mixed	1,142: 378 / 215 / 529 / 20
antiparallel hairpins / proteins with an antiparallel pair	104 / 120
sequences read back exactly from their sequence circuits	150 of 150
same prime and gate counts as the project's Python campaign	150 and 150 of 150

The 1fnaA example (91 residues, $p = 7$, $2^{14} = 16,384$ inputs): 201 contacts, 216 gates, 111 rules (16 block rules holding 56 contacts), exact on every input. One gate, read as a rule: $i = 001101-$, $j = 100001-$ (low bit free in each field), i.e. IF residue 27–28 AND residue 67–68 THEN contact, with no exception. Its strand pairs (1-based residues; for antiparallel pairs the register $i + j$ is given in 1-based numbering):

segment	segment	contacts	block rules	direction	register
25–34	61–70	22	6	antiparallel	$i + j = 94-98$
1–8	14–18	14	4	antiparallel (hairpin)	$i + j = 18-22$
11–14	53–56	10	3	antiparallel	$i + j = 65-68$
57–60	89–90	6	2	parallel	$j - i = 29-32$
9–10	89–90	4	1	short	–

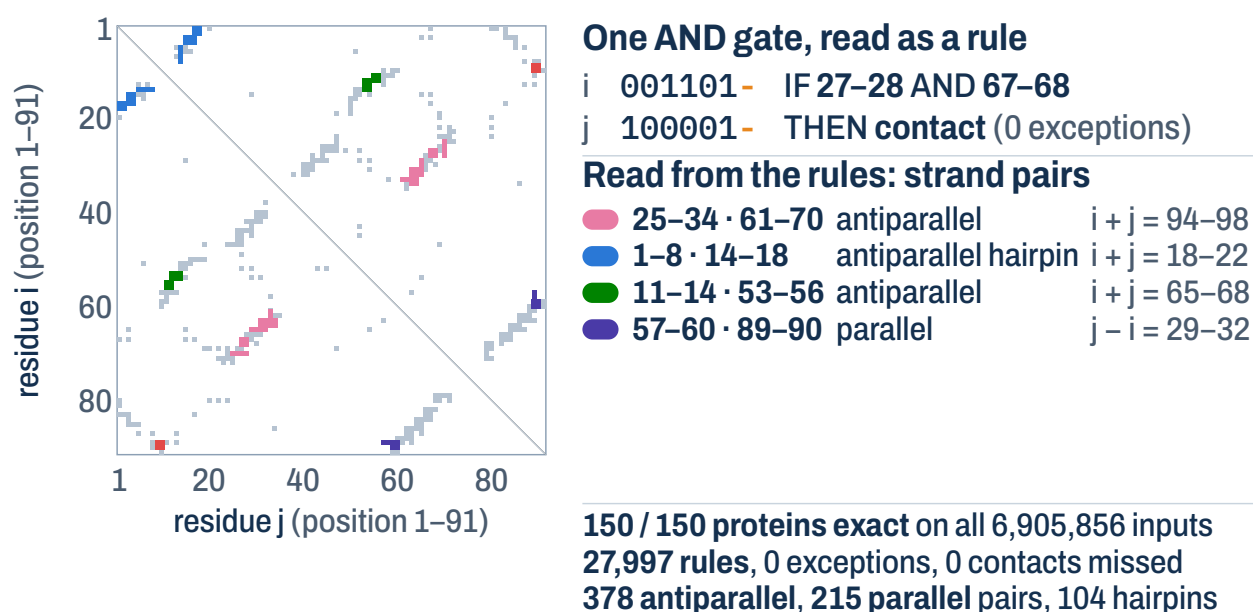


Figure 12. Band 100, left: the contact map of 1fnaA (rows residue i , columns residue j , 1–91) with its block rules coloured by strand pair, one AND gate read as a rule, the strand pairs read from the rules, and the dataset totals.

11.4 121 high-resolution chains: protein, DNA and RNA

The same pipeline on a curated set of 106 X-ray structures at 0.48–1.39 Å resolution (protein, DNA and RNA). They give 154 chains (a protein chain, or the 2–4 strands of a nucleic acid taken together); 33 have no contact under the definition, which leaves 121. Nucleic acids use C1' atoms (P if missing), 10 Å and a separation of at least 3 along a strand; any two strands may touch.

Table 7. Curated set by type: all circuits exact, no rule with an exception, no contact missed.

type	chains	contacts	inputs	gates	rules	exceptions	anti / par / hairpins	base pairs WC
protein	81	27,562	5,971,968	31,956	16,563	0	216 / 76 / 68	–
DNA	25	499	29,440	674	355	0	1 / 0 / 0	231 of 231
RNA	15	63	2,688	96	48	0	0 / 0 / 0	12 of 15

Algorithm 10 Base pairs by geometry (nucleic acids)

- 1: **for** each pair of nucleotides $i < j$ **do**
- 2: **if** same strand **and** sequence numbers differ by < 4 **then continue** ▷ at least 3 unpaired between
- 3: **if** one is a purine and the other a pyrimidine **and** purine N1 to pyrimidine N3 < 3.5 Å **then**
- 4: record a base pair; Watson–Crick if A–T/U or G–C, wobble if G–U

In the DNA chains all 231 base pairs found this way are Watson–Crick; in the RNA chains 12 of 15 are.

11.5 One species: 1,299 SARS-CoV-2 Omicron Spike sequences

Data. 1,299 Spike protein sequences (301 distinct), aligned (1,276 columns). Every sequence carries at least two Omicron-defining Spike substitutions of the twelve we checked (1,225 carry all twelve). Columns are numbered as in Wuhan-Hu-1 (UniProt P0DTC2) by a global alignment of the column consensus to the reference (consensus identity 0.978). The co-evolving column pairs come from the project's co-evolution campaign: columns with Shannon entropy above 0.3, pairs at most 30 columns apart with mutual information above 0.1 and at least 5 mutated sequences; the 10 strongest pairs are analysed.

Rules, all counted over every sequence. For a column pair (p, q) , using the sequences with a standard residue at both columns:

- *allowed* combinations: every residue pair that occurs;
- *forbidden* combinations: residues that each occur in their own column but never together;
- *implications* “X at $p \Rightarrow Y$ at q ” (and the reverse): every sequence with X at p has Y at q , with at least 5 such sequences.

Each pair is also minimised as two circuits, one whose 1-cells are the allowed combinations and one whose 1-cells are the forbidden ones; in every case the cover consists of essential primes only, so it is the *unique minimum* cover.

Table 8. All ten co-evolving position pairs (Wuhan-Hu-1 numbering; “ins.” is an inserted column with no reference residue, between 212 and 213). n : sequences with standard residues at both positions. Counts in parentheses.

pair	n	allowed (sequences)	forbidden	implications (support, 0 counterexamples)
S371 $\times$ T376	1,294	FA (970), LT (311), SA (1), ST (12)	FT, LA	F371 $\Rightarrow$ A376 (970); L371 $\Rightarrow$ T376 (311)
T19 $\times$ A27	1,276	IS (957), TA (318), TS (1)	IA	I19 $\Rightarrow$ S27 (957); A27 $\Rightarrow$ T19 (318)
T376 $\times$ D405	1,297	AD (3), AN (970), TD (324)	TN	T376 $\Rightarrow$ D405 (324); N405 $\Rightarrow$ A376 (970)
A67 $\times$ T95	1,298	AI (1), AT (983), VI (314)	VT	V67 $\Rightarrow$ I95 (314); T95 $\Rightarrow$ A67 (983)
D405 $\times$ R408	1,289	DR (319), NR (12), NS (958)	DS	D405 $\Rightarrow$ R408 (319); S408 $\Rightarrow$ N405 (958)
N211 $\times$ R214	1,259	IE (271), IK (1), IR (1), KR (1), NR (985)	KE, KK, NE, NK	N211 $\Rightarrow$ R214 (985); E214 $\Rightarrow$ I211 (271)
N211 $\times$ V213	1,258	IP (272), KG (1), NE (1), NG (976), NV (8)	IE, IG, IV, KE, KP, KV, NP	I211 $\Rightarrow$ P213 (272); P213 $\Rightarrow$ I211 (272); V213 $\Rightarrow$ N211 (8)
V213 $\times$ R214	1,278	ER (1), GR (977), PE (278), PK (1), VR (21)	EE, EK, GE, GK, PR, VE, VK	G213 $\Rightarrow$ R214 (977); V213 $\Rightarrow$ R214 (21); E214 $\Rightarrow$ P213 (278)
ins. $\times$ R214	301	GR (1), IR (13), LR (6), SR (2), VE (278), VK (1)	GE, GK, IE, IK, LE, LK, SE, SK, VR	I(ins.) $\Rightarrow$ R214 (13); L(ins.) $\Rightarrow$ R214 (6); E214 $\Rightarrow$ V(ins.) (278)
F486 $\times$ Q493	1,297	FQ (20), FR (1,173), PQ (2), VQ (102)	PR, VR	V486 $\Rightarrow$ Q493 (102); R493 $\Rightarrow$ F486 (1,173)

Totals: 10 position pairs, 41 allowed and 35 forbidden combinations, 23 implications, 0 violations: every one of the 1,299 sequences obeys every rule. Every sequence’s own sequence circuit reads it back exactly.

Are the rule pairs structural contacts? On the Omicron BA.2 receptor-binding domain (PDB 7XB0, chain B, residues 333–527; 482 contacts under the protein definition) the four rule pairs inside the RBD are not contacts:

pair	residues in 7XB0	separation	$C\beta$ distance (Å)	contact
371–376	F371, A376	5	14.52	no
376–405	A376, N405	29	12.78	no
405–408	N405, S408	3	7.05	no
486–493	F486, R493	7	17.46	no

The structure carries the most common combination at each pair. The rules record which residues are inherited together in this lineage, not residues that touch.

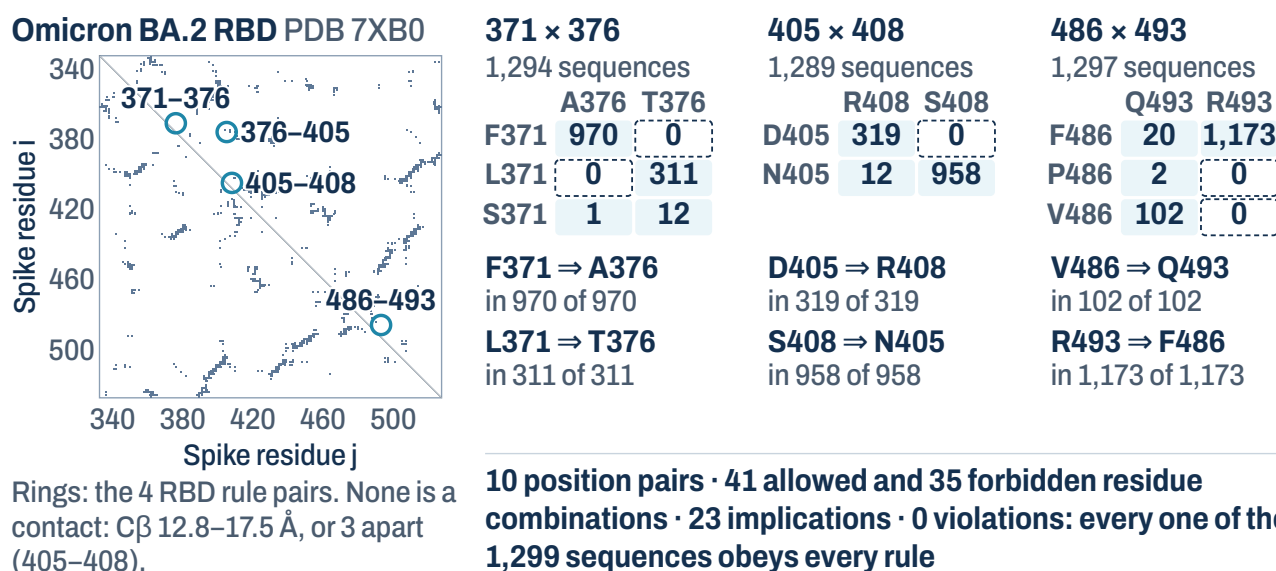


Figure 13. Band 100, middle: the Omicron BA.2 RBD contact map with the four rule pairs ringed, three of the residue-pair tables (a dashed 0 is a forbidden combination), their implications and the totals.

12 Takeaways

- A verified foundation.** The encodings are Lean theorems (Appendix A); every circuit is exact on all its inputs (150 of 150 PSICOV proteins, 121 of 121 curated chains).
- Rules that are true, not likely.** Each holds for every contact, sequence or frame it names: 0 exceptions among 27,997 PSICOV rules, 0 violations in 1,299 Spike sequences, 0 counterexample frames.
- One language for sequence, structure and dynamics.** The same encoding and the same minimiser read k -mer maps, contact maps, co-evolving positions and contact states along folding trajectories.
- Lossless by proof.** Each cell decodes to both residues (T124), and Gray order only permutes cells (T127, T131), so no statistic changes.
- Exact about absence.** No rule on residue chemistry alone holds without exception (algorithm below).

Algorithm 11 Is there an exception-free rule on chemistry alone?

- groups: AVLI, FYW, MC, PG, STNQ, DE, HKR; separation bins σ : 6–10, 11–20, 21–30, ≥ 31
 - for** each PSICOV protein and each residue pair $i < j$ with $j - i \geq 6$ **do**
 - add the pair to the class (group of i , group of j , $\sigma(j - i)$), as a contact or a non-contact
 - a rule “class $\Rightarrow$ contact” or “class $\Rightarrow$ no contact” is exception-free iff its class is pure
-

Result: 1,675,886 residue pairs (47,430 contacts) fall into all $7 \times 7 \times 4 = 196$ classes, and 196 of 196 classes contain both contacts and non-contacts. Exact rules come from each structure’s own circuit, not from chemistry and separation alone.

13 The demo website

e-motioner-x-sbs.github.io/ICMS_Submission runs the whole pipeline in the browser on any PDB entry (or on five bundled examples that work offline: 1FNA, 1UBQ, 2CI2, 1BNA, 1EHZ), in thirteen chapters numbered with a 4-bit Gray code:

0000	Pick a structure	0101	Minimisation finds blocks
0001	Letters become bits	0100	The structure as a circuit
0011	A code that changes one bit at a time	1100	From the circuit, rules
0010	A Karnaugh map of your sequence	1101	From the rules, inferences
0110	The geometry of the code	1111	Read the sequence back
0111	Contacts become a Boolean function	1110	What the code can and cannot see
		1010	What it all means

Every chapter shows the theorems behind its step as mathematical statements and can re-run their finite checks; the minimiser runs in a background worker; the finished circuit can be run on every input with one button and downloaded as a PLA (Espresso) or Verilog file, so any logic tool can re-verify it. The poster's QR code opens the site.

14 Presenting the poster

A two-minute walk-through

1. *000*. “People draw sequences on Gray-coded grids and trust their properties. We proved them in Lean: 131 theorems. Follow 1fnaA: residues become 5-bit codes, a pair becomes one cell, neighbouring cells differ in one bit.”
2. *001–010*. “Gray order changes one bit per step; DNA's A, C, G, T is already a Gray cycle; 20 residues on a 5-cube, ordered by chemistry; all 190 distances proved.”
3. *110*. “A contact map is a Boolean function. Quine–McCluskey merges one-bit neighbours into cubes; a cover of prime cubes is a circuit equal to the map, checked on every input.”
4. *101*. “The proofs guarantee three things we use: rules never fire on padding, a cube is a chain segment, and Gray coding only permutes cells.”
5. *100*. “Read the gates as rules: 150 proteins, 27,997 rules, no exception, and the block rules give strand pairs and their direction by counting. For a species, 1,299 Omicron Spike sequences: 35 forbidden combinations and 23 implications, no violation.”

Questions you may be asked

Is the circuit minimal?

Exact, yes (sound and complete, checked on every input). Minimal, not in general: the cover is all essential primes plus a greedy completion. It is the unique minimum cover where it has essential primes only, as for every Spike circuit.

What does “compiled check” trust?

Lean's compiler, in addition to its kernel: 100 of the 131 theorems are proved by evaluating every case in compiled code; 23 are checked by the kernel with standard axioms, 8 with no axioms at all.

Why Gray code and not plain binary?

Gray order makes touching cells differ in one variable, the property Karnaugh maps need. It does not change any statistic of the cell values (it permutes cells); it changes which cells touch. For contact maps the position bits are plain binary, because then a cube is a chain segment (T119); Gray-coded positions also give segments, but different ones (T122, T123).

Are the Spike rules causal, or contacts?

Neither is claimed. They are exact statements about this set of sequences; the four RBD rule pairs are not structural contacts ($C\beta$ 12.8–17.5 Å, or 3 residues apart).

Do the rules predict contacts for a new protein?

No. Each circuit is exact for its own structure. Chemistry and separation alone give no exception-free

rule: all 196 classes are mixed (§12).

Why do some strand pairs have no direction?

A single block is a rectangle and has no direction (“short”, 529 of 1,142); when both spreads are equal the pair is “mixed” (20).

Is the Lean code available?

It is private while the work is under review. Every statement is in Appendix A and can be re-checked in the browser.

15 Data, code and reproducibility

Every number in this document is produced by a script and stored in a result file:

script	what it computes
poster/data/exact_psicov.mjs	PSICOV circuits, exhaustive checks, rules, inferences, sequence read-back
poster/data/exact_dataset.mjs	the curated protein/DNA/RNA set, including base pairs
poster/data/prep_spike_rules.py	Spike allowed/forbidden combinations, implications, Wuhan numbering
poster/data/prep_spike_rbd.py	RBD contact map of 7XB0 and the rule-pair distances
poster/data/prep_sbm_exact.py	every-frame rules of the folding trajectories
poster/data/prep_absence.py	the 196 chemistry × separation classes
poster/data/summarise_exact.mjs	one summary file read by the poster and the site
docs/js/core/qm.js, rules.js	the minimiser, rules and inferences (shared by poster and site)

The browser engine reproduces the project’s Python campaign: identical prime and gate counts for all 150 PSICOV proteins. For 10 structures, the circuits exported by the website were also re-checked independently, against contact maps computed with a separate structure reader (gemmi): all agree on every cell.

References

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Structures: PDB entries 1FNA, 1BNA, 1CBN, 7XB0. Reference sequence: UniProt P0DTC2.

A All 131 theorems, as mathematical statements

Each theorem is stated in ordinary notation; the statement is exactly what the Lean theorem proves. After each: its meaning in words, how many cases the browser re-check evaluates, and (top right) what the proof trusts. Numbers T1–T131 are used throughout this document.

Notation used in the statements

$$[n] = \{0, 1, \dots, n-1\}$$

$\oplus$ is bitwise exclusive or

$g(n) = n \oplus \lfloor n/2 \rfloor$ is the Gray code

$\bar{g}(m) = \bigoplus_{k=0}^7 \lfloor m/2^k \rfloor$ is the 8-bit Gray decoder

$d(a, b)$ is the number of bits in which a and b differ

$2^{\mathbb{N}} = \{1, 2, 4, 8, \dots\}$, so $a \oplus b \in 2^{\mathbb{N}}$ says that a and b differ in exactly one bit

$\mathcal{N} = \{A, C, G, T\}$, with codes $e(A) = 0, e(C) = 1, e(G) = 3, e(T) = 2$ (binary 00, 01, 11, 10)

$r(x)$ and $c(x)$ are the row and column of letter x on the 2×2 map: $r(A) = r(C) = 0, r(G) = r(T) = 1, c(A) = c(T) = 0, c(C) = c(G) = 1$

$E_k(x_1 \dots x_k) = \sum_{i=1}^k 4^{k-i} e(x_i)$ is the cell of a k -mer; $R(xy) = e(x)$ and $C(xy) = e(y)$ are the row and column of a dinucleotide

$\mathcal{A}$ is the 20 amino acids in group order A V L I F Y W M C P G S T N Q D E H K R; $e(a) \in [20]$ is the position of a in that order and $\gamma(a) = g(e(a))$ its 5-bit code

$\delta(a, b) = d(\gamma(a), \gamma(b))$ is the distance between the codes of amino acids a and b

$\rho(a) = a$ is the raw group-order index of amino acid number $a \in [20]$

C_8 is the 8-residue example map: $C_8(i, j) = 1$ for $(i, j) = (0, 3), (1, 4), (2, 5), (0, 5)$ and their mirror images, 0 otherwise

$c_{ij} = 8g(i) + g(j)$ is the Gray-indexed cell of the residue pair (i, j)

$P = \{c_{03}, c_{14}, c_{25}, c_{05}\}$ is the set of cells holding the example's contacts

$N_k = \#\{(a, b) \in [20]^2 \mid a \neq b, d(g(a), g(b)) = k\}$ counts the ordered pairs of amino-acid codes at distance k

$\sigma(s)$ is the separation bin of residues s apart: 0 for $s \leq 10$, 1 for $11 \leq s \leq 20$, 2 for $21 \leq s \leq 30$, 3 for $s \geq 31$

$Q_w(v, m)$ is the set of $c \in [2^w]$ that agree with v on every bit not set in m : the cube with fixed bits v and free bits m

T is a worked table on two bits, $T(0) = T(1) = 1$ and $T(2) = T(3) = 0$, covered by the single cube $Q_2(0, 1)$

$2^u \mid v$ says that v is a multiple of 2^u

A.1 Gray code and the DNA letters (16 theorems)

T1

compiled check of every case

$$\forall n \in [256] : g(n) < 256$$

Every 8-bit number keeps an 8-bit Gray code: for n from 0 to 255, its Gray code is below 256. (re-checked on all 256 cases.)

T2

compiled check of every case

$$\forall n \in [255] : g(n) \oplus g(n+1) \in 2^{\mathbb{N}}$$

Counting from 0 to 255 in Gray code changes exactly one bit per step: the codes of n and $n+1$ differ in a single bit. (re-checked on all 255 cases.)

T3

compiled check of every case

$$\forall n \in [255] : d(g(n), g(n+1)) = 1$$

The same fact measured in bits: consecutive Gray codes are at Hamming distance exactly 1, for every n below 255. (re-checked on all 255 cases.)

T4

compiled check of every case

$$\forall a, b \in [256] : g(a) = g(b) \Rightarrow a = b$$

No two 8-bit numbers share a Gray code, so nothing is lost by recoding. (re-checked on all 65,536 cases.)

T5

compiled check of every case

$$\forall n \in [256] : \bar{g}(g(n)) = n$$

Decoding undoes encoding, for every 8-bit number. (re-checked on all 256 cases.)

T6

kernel-checked, standard axioms

$$\forall x, y \in \mathcal{N} : e(x) = e(y) \Rightarrow x = y$$

The four DNA letters get four different 2-bit codes (A=00, C=01, G=11, T=10). (re-checked on all 16 cases.)

T7

compiled check of every case

$$g(e(A)) \oplus g(e(T)) = 3$$

Watson–Crick partners A and T differ in both bits of their Gray-coded labels: their codes XOR to 11. (a closed statement, evaluated once.)

T8

compiled check of every case

$$g(e(C)) \oplus g(e(G)) = 3$$

Watson–Crick partners C and G differ in both bits of their Gray-coded labels: their codes XOR to 11. (a closed statement, evaluated once.)

T9

compiled check of every case

$$e(A) \oplus e(C) \in 2^{\mathbb{N}}$$

A (00) and C (01) differ in one bit: the first step of the Gray cycle $A \rightarrow C \rightarrow G \rightarrow T \rightarrow A$. (a closed statement, evaluated once.)

T10

compiled check of every case

$$e(C) \oplus e(G) \in 2^{\mathbb{N}}$$

C (01) and G (11) differ in one bit. (a closed statement, evaluated once.)

T11

compiled check of every case

$$e(G) \oplus e(T) \in 2^{\mathbb{N}}$$

G (11) and T (10) differ in one bit. (a closed statement, evaluated once.)

T12

compiled check of every case

$$e(T) \oplus e(A) \in 2^{\mathbb{N}}$$

T (10) and A (00) differ in one bit, closing the cycle. (a closed statement, evaluated once.)

T13

compiled check of every case

$$g(e(A)) \oplus g(e(G)) \in 2^{\mathbb{N}}$$

With Gray-coded labels ($A = 00$, $G = 10$) the purine–purine transition $A \leftrightarrow G$ flips one bit. (a closed statement, evaluated once.)

T14

compiled check of every case

$$g(e(C)) \oplus g(e(T)) \in 2^{\mathbb{N}}$$

With Gray-coded labels ($C = 01$, $T = 11$) the pyrimidine–pyrimidine transition $C \leftrightarrow T$ flips one bit. (a closed statement, evaluated once.)

T15

compiled check of every case

$$d(g(e(A)), g(e(C))) = 1$$

With Gray-coded labels the transversion $A \leftrightarrow C$ is also one bit (00 vs 01). (a closed statement, evaluated once.)

T16

compiled check of every case

$$e(A) \neq e(C), e(A) \neq e(G), e(A) \neq e(T), e(C) \neq e(G), e(C) \neq e(T), e(G) \neq e(T)$$

All six pairs of DNA codes are different. (a closed statement, evaluated once.)

A.2 k-mer Karnaugh maps (35 theorems)

T17

compiled check of every case

$$\forall n \in [4] : g(n) < 4$$

On two bits the Gray code stays on two bits: $g(n) < 4$ for $n < 4$. (re-checked on all 4 cases.)

T18

compiled check of every case

$$\forall a, b \in [4] : g(a) = g(b) \Rightarrow a = b$$

On two bits, different numbers get different Gray codes. (re-checked on all 16 cases.)

T19

compiled check of every case

$$\forall n \in [3] : g(n) \oplus g(n+1) \in 2^{\mathbb{N}}$$

Counting 0, 1, 2, 3 in 2-bit Gray code flips one bit per step. (re-checked on all 3 cases.)

T20

compiled check of every case

$$g(3) \oplus g(0) \in 2^{\mathbb{N}}$$

The 2-bit Gray cycle closes: the codes of 3 and 0 also differ in one bit, so K-map edges wrap around. (a closed statement, evaluated once.)

T21

kernel-checked, standard axioms

$$\forall x, y \in \mathcal{N} : e(x) = e(y) \Rightarrow x = y$$

The four DNA letters get four different codes (the same encoding, restated for the k-mer maps). (re-checked on all 16 cases.)

T22

kernel-checked, standard axioms

$$\forall x, y \in \mathcal{N} : e(x) = e(y) \Rightarrow x = y$$

Comparing the codes as plain numbers is enough to tell the letters apart. (re-checked on all 16 cases.)

T23

compiled check of every case

$$e(A) \oplus e(C) \in 2^{\mathbb{N}}$$

A (00) and C (01) differ in one bit (restated for the k-mer maps). (a closed statement, evaluated once.)

T24

compiled check of every case

$$e(C) \oplus e(G) \in 2^{\mathbb{N}}$$

C (01) and G (11) differ in one bit. (a closed statement, evaluated once.)

T25

compiled check of every case

$$e(G) \oplus e(T) \in 2^{\mathbb{N}}$$

G (11) and T (10) differ in one bit. (a closed statement, evaluated once.)

T26

compiled check of every case

$$e(T) \oplus e(A) \in 2^{\mathbb{N}}$$

T (10) and A (00) differ in one bit. (a closed statement, evaluated once.)

T27

compiled check of every case

$$g(e(A)) \oplus g(e(G)) \in 2^{\mathbb{N}}$$

With Gray-coded labels, $A \leftrightarrow G$ (a transition) flips one bit. (a closed statement, evaluated once.)**T28**

compiled check of every case

$$g(e(C)) \oplus g(e(T)) \in 2^{\mathbb{N}}$$

With Gray-coded labels, $C \leftrightarrow T$ (a transition) flips one bit. (a closed statement, evaluated once.)**T29**

compiled check of every case

$$g(e(A)) \oplus g(e(T)) = 3$$

With Gray-coded labels, A and T XOR to 11, both bits differ. (a closed statement, evaluated once.)

T30

compiled check of every case

$$g(e(C)) \oplus g(e(G)) = 3$$

With Gray-coded labels, C and G XOR to 11, both bits differ. (a closed statement, evaluated once.)

T31

kernel-checked, standard axioms

$$\forall x, y \in \mathcal{N} : 2r(x) + c(x) = 2r(y) + c(y) \Rightarrow x = y$$

On the 2×2 map every nucleotide gets its own cell. (re-checked on all 16 cases.)

T32

kernel-checked, standard axioms

$$\forall x, y \in \mathcal{N} : r(x) = r(y) \wedge c(x) = c(y) \Rightarrow x = y$$

A row and a column together pin down exactly one nucleotide. (re-checked on all 16 cases.)

T33

kernel-checked, no axioms

$$r(A) = r(C) \wedge c(A) + 1 = c(C)$$

A and C sit side by side in the same row of the 2×2 map. (a closed statement, evaluated once.)

T34

kernel-checked, no axioms

$$r(C) + 1 = r(G) \wedge c(C) = c(G)$$

C and G sit one above the other in the same column. (a closed statement, evaluated once.)

T35

kernel-checked, no axioms

$$r(G) = r(T) \wedge c(G) = c(T) + 1$$

G and T sit side by side in the same row. (a closed statement, evaluated once.)

T36

kernel-checked, no axioms

$$r(T) = r(A) + 1 \wedge c(T) = c(A)$$

T and A sit one above the other, so the cycle $A \rightarrow C \rightarrow G \rightarrow T \rightarrow A$ walks round the 2×2 map. (a closed statement, evaluated once.)

T37

kernel-checked, standard axioms

$$\forall xy \in \mathcal{N}^2 : E_2(xy) = 4R(xy) + C(xy)$$

A dinucleotide's cell is $\text{row} \times 4 + \text{column}$, with the first letter giving the row and the second the column. (re-checked on all 16 cases.)

T38

kernel-checked, standard axioms

$$\forall u, v \in \mathcal{N}^2 : E_2(u) = E_2(v) \Rightarrow u = v$$

The 16 dinucleotides land in 16 different cells of the 4×4 map. (re-checked on all 256 cases.)

T39

kernel-checked, standard axioms

$$\forall u \in \mathcal{N}^2 : E_2(u) < 16$$

Every dinucleotide cell index is below 16: the 4×4 map has no overflow. (re-checked on all 16 cases.)

T40

kernel-checked, no axioms

$$\forall u \in \mathcal{N}^2 : R(u) < 4$$

Every dinucleotide row index is below 4. (re-checked on all 16 cases.)

T41

kernel-checked, no axioms

$$\forall u \in \mathcal{N}^2 : C(u) < 4$$

Every dinucleotide column index is below 4. (re-checked on all 16 cases.)

T42

compiled check of every case

$$\forall x \in \mathcal{N} : R(Cx) - R(Ax) = 1, R(Gx) - R(Cx) = 2, R(Ax) + 4 - R(Tx) = 2$$

Stepping the first letter $A \rightarrow C \rightarrow G \rightarrow T$ moves the raw row index by +1, +2 and (wrapping) 2: raw numbers are not Gray distances, which is why the map orders its axes by the Gray cycle. (re-checked on all 4 cases.)

T43

compiled check of every case

$$\forall x \in \mathcal{N} : C(xC) - C(xA) = 1, C(xG) - C(xC) = 2, C(xA) + 4 - C(xT) = 2$$

The same for the second letter and the column index. (re-checked on all 4 cases.)

T44

kernel-checked, standard axioms

$$\forall u, v \in \mathcal{N}^2 : R(u) = R(v) \wedge C(u) = C(v) \Rightarrow u = v$$

Knowing a dinucleotide's row and column tells you the dinucleotide. (re-checked on all 256 cases.)

T45

kernel-checked, standard axioms

$$\forall u, v \in \mathcal{N}^3 : E_3(u) = E_3(v) \Rightarrow u = v$$

The 64 trinucleotides land in 64 different cells of the 8×8 map. (re-checked on all 4,096 cases.)

T46

kernel-checked, standard axioms

$$\forall t \in \mathcal{N}^3 : E_3(t) < 64$$

Every trinucleotide cell index is below 64. (re-checked on all 64 cases.)

T47

kernel-checked, standard axioms

$$\forall u, v \in \mathcal{N}^4 : E_4(u) = E_4(v) \Rightarrow u = v$$

The 256 tetranucleotides land in 256 different cells of the 16×16 map. (re-checked on all 65,536 cases.)

T48

kernel-checked, standard axioms

$$\forall t \in \mathcal{N}^4 : E_4(t) < 256$$

Every tetranucleotide cell index is below 256. (re-checked on all 256 cases.)

T49

compiled check of every case

$$\forall x, y \in \mathcal{N}, x \neq y : 1 \leq d(g(e(x)), g(e(y))) \leq 2$$

Any single-letter substitution changes one or two bits of the Gray-coded label, never zero and never more than two. (re-checked on all 16 cases.)

T50

compiled check of every case

$$d(g(e(A)), g(e(T))) = 2 \wedge d(g(e(C)), g(e(G))) = 2$$

Watson–Crick partners are as far apart as two bits allow: A–T and C–G are both at distance 2 with Gray-coded labels. (a closed statement, evaluated once.)

T51

compiled check of every case

$$d(g(e(A)), g(e(G))) = 1 \wedge d(g(e(C)), g(e(T))) = 1$$

Transitions A–G and C–T are both at distance 1 with Gray-coded labels. (a closed statement, evaluated once.)

A.3 The amino-acid code (27 theorems)

T52

kernel-checked, standard axioms

$$\forall a, b \in \mathcal{A} : e(a) = e(b) \Rightarrow a = b$$

The 20 amino acids get 20 different 5-bit codes. (re-checked on all 400 cases.)

T53

kernel-checked, standard axioms

$$\forall a \in \mathcal{A} : e(a) < 32$$

Every amino-acid index fits in five bits (below 32). (re-checked on all 20 cases.)

T54

compiled check of every case

$$\delta(A, V), \delta(A, L), \delta(A, I), \delta(V, L), \delta(V, I), \delta(L, I) \leq 2$$

The aliphatic hydrophobics A, V, L, I are all within two bits of each other. (a closed statement, evaluated once.)

T55

compiled check of every case

$$\delta(F, Y), \delta(F, W), \delta(Y, W) \leq 2$$

The aromatics F, Y, W are all within two bits of each other. (a closed statement, evaluated once.)

T56

compiled check of every case

$$\delta(M, C) = 1$$

The sulfur pair M and C is one bit apart. (a closed statement, evaluated once.)

T57

compiled check of every case

$$\delta(P, G) = 1$$

The structure breakers P and G are one bit apart. (a closed statement, evaluated once.)

T58

compiled check of every case

$$\delta(S, T), \delta(S, N), \delta(S, Q), \delta(T, N), \delta(T, Q), \delta(N, Q) \leq 3$$

The polar uncharged S, T, N, Q are within three bits of each other (S–Q = 3 is the widest group). (a closed statement, evaluated once.)

T59	compiled check of every case
$\delta(D, E) = 1$	
The acids D and E are one bit apart. (a closed statement, evaluated once.)	
T60	compiled check of every case
$\delta(H, K), \delta(H, R), \delta(K, R) \leq 2$	
The bases H, K, R are within two bits of each other. (a closed statement, evaluated once.)	
T61	compiled check of every case
$\delta(D, E) = 1$	
D and E: distance 1. (a closed statement, evaluated once.)	
T62	compiled check of every case
$\delta(H, K) = 1$	
H and K: distance 1. (a closed statement, evaluated once.)	
T63	compiled check of every case
$\delta(K, R) = 1$	
K and R: distance 1. (a closed statement, evaluated once.)	
T64	compiled check of every case
$\delta(H, R) = 2$	
H and R: distance 2 (they meet through K). (a closed statement, evaluated once.)	
T65	compiled check of every case
$\delta(S, T) = \delta(T, N) = \delta(N, Q) = 1$	
S → T → N → Q is a path of one-bit steps. (a closed statement, evaluated once.)	
T66	compiled check of every case
$\delta(S, Q) = 3$	
The two ends of that path, S and Q, are three bits apart. (a closed statement, evaluated once.)	
T67	compiled check of every case
$\delta(F, M) = 1, \delta(F, C) = 2, \delta(Y, M) = 2, \delta(Y, C) = 3, \delta(W, M) = 1, \delta(W, C) = 2$	
All six aromatic-sulfur distances: F-M 1, F-C 2, Y-M 2, Y-C 3, W-M 1, W-C 2. (a closed statement, evaluated once.)	
T68	compiled check of every case
$\delta(F, H) = 5$	
F and H differ in all five bits. (a closed statement, evaluated once.)	
T69	compiled check of every case
$\delta(Y, E) = 5$	
Y and E differ in all five bits. (a closed statement, evaluated once.)	
T70	compiled check of every case
$\delta(W, R) = 5$	
W and R differ in all five bits. (a closed statement, evaluated once.)	
T71	compiled check of every case
$\delta(M, K) = 5$	
M and K differ in all five bits. (a closed statement, evaluated once.)	

T72

compiled check of every case

$$\delta(A, K) \geq 4, \delta(V, H) \geq 2, \delta(L, R) \geq 3$$

Hydrophobics sit away from the bases: A–K ≥ 4 , V–H ≥ 2 , L–R ≥ 3 . (a closed statement, evaluated once.)**T73**

compiled check of every case

$$\delta(A, D) = 1, \delta(V, E) = 3, \delta(I, D) = 2$$

A known weak spot: A and D are one bit apart; V–E is 3 and I–D is 2. (a closed statement, evaluated once.)

T74

compiled check of every case

$$\delta(a, b) = 1 \text{ for } (a, b) = \\ (A, V), (A, I), (V, L), (L, I), (F, Y), \\ (Y, W), (M, C), (P, G), (S, T), (T, N), \\ (N, Q), (D, E), (H, K), (K, R)$$

The 14 one-bit pairs that stay inside a chemical group. (a closed statement, evaluated once.)

T75

compiled check of every case

$$\delta(a, b) = 1 \text{ for } (a, b) = (A, M), (A, D), (V, W), (V, Q), (L, Y), (L, N), (I, F), (I, T)$$

Eight one-bit pairs that cross from the aliphatic group into another group. (a closed statement, evaluated once.)

T76

compiled check of every case

$$\delta(a, b) = 1 \text{ for } (a, b) = (F, M), (F, S), (Y, G), (W, M), (W, P)$$

Five one-bit pairs that cross from the aromatic group into another group. (a closed statement, evaluated once.)

T77

compiled check of every case

$$\delta(a, b) = 1 \text{ for } (a, b) = (C, P), (C, S), (C, D), (E, H), (E, R)$$

Five more cross-group one-bit pairs, including E–H and E–R between opposite charges. (a closed statement, evaluated once.)

T78

compiled check of every case

$$\delta(a, b) = 1 \text{ for } (a, b) = (S, G), (G, N), (P, Q), (T, D), (T, R), (N, K), (Q, D), (Q, H)$$

The last eight cross-group one-bit pairs, all touching the polar or structure-breaking groups. (a closed statement, evaluated once.)

A.4 An 8-residue contact map (20 theorems)**T79**

compiled check of every case

$$\forall i, j \in [8] : C_8(i, j) = C_8(j, i)$$

In the fixed 8-residue example, i touches j exactly when j touches i. (re-checked on all 64 cases.)

T80

compiled check of every case

$$\forall i \in [8] : C_8(i, i) = 0$$

In the 8-residue example no residue is in contact with itself. (re-checked on all 8 cases.)

T81

compiled check of every case

$$C_8(0, 3) = C_8(1, 4) = C_8(2, 5) = C_8(0, 5) = 1$$

The example's four contacts are (0,3), (1,4), (2,5) and (0,5). (a closed statement, evaluated once.)

T82

kernel-checked, standard axioms

$$\forall i, j \in [8] : 8i + j < 64$$

Cell $i \times 8 + j$ of an 8×8 contact map is below 64. (re-checked on all 64 cases.)

T83

kernel-checked, standard axioms

$$\forall i_1, j_1, i_2, j_2 \in [8] : 8i_1 + j_1 = 8i_2 + j_2 \Rightarrow i_1 = i_2 \wedge j_1 = j_2$$

Different residue pairs get different cells of the 8×8 map. (re-checked on all 4,096 cases.)

T84

kernel-checked, standard axioms

$$\forall i, j \in [8], i \neq j : 8i + j \neq 8j + i$$

A contact (i, j) and its mirror (j, i) occupy two different cells whenever $i \neq j$. (re-checked on all 64 cases.)**T85**

compiled check of every case

$$\forall i \in [8] : g(i) < 8$$

The Gray code of a residue index below 8 stays below 8. (re-checked on all 8 cases.)

T86

compiled check of every case

$$\forall i, j \in [8] : g(i) = g(j) \Rightarrow i = j$$

Gray-coding residue indices 0–7 loses nothing. (re-checked on all 64 cases.)

T87

compiled check of every case

$$\forall i, j \in [8] : 8g(i) + g(j) < 64$$

The Gray-indexed cell of an 8×8 contact map is below 64. (re-checked on all 64 cases.)

T88

compiled check of every case

$$\forall i_1, j_1, i_2, j_2 \in [8] : 8g(i_1) + g(j_1) = 8g(i_2) + g(j_2) \Rightarrow i_1 = i_2 \wedge j_1 = j_2$$

Gray-indexed cells are also one per residue pair. (re-checked on all 4,096 cases.)

T89

compiled check of every case

$$\exists i, j \in [8] : 8g(i) + g(j) \neq 8i + j$$

Gray indexing really moves cells: the pair (1, 2) lands in a different cell than under plain indexing. (an existence statement: the check finds a witness.)

T90

compiled check of every case

$$c_{03} = 2, c_{14} = 14, c_{25} = 31, c_{05} = 7$$

The example's four contacts land in Gray cells 2, 14, 31 and 7. (a closed statement, evaluated once.)

T91

compiled check of every case

$$c_{03} \neq c_{30}, c_{14} \neq c_{41}, c_{25} \neq c_{52}, c_{05} \neq c_{50}$$

Each of those contacts and its mirror image are two different Gray cells. (a closed statement, evaluated once.)

T92

compiled check of every case

$$C_8(i, j) = 1 \text{ for } (i, j) = (0, 3), (3, 0), (1, 4), (4, 1), (2, 5), (5, 2), (0, 5), (5, 0)$$

Read as a Boolean function, the example is 1 on eight cells: the four contacts and their mirrors. (a closed statement, evaluated once.)

T93

compiled check of every case

$$\forall i, j \in [8] : C_8(i, j) = 1 \Leftrightarrow (i, j) \in \{(0, 3), (3, 0), (1, 4), (4, 1), (2, 5), (5, 2), (0, 5), (5, 0)\}$$

...and 0 on all other 56: the function is fully specified. (re-checked on all 64 cases.)

T94

compiled check of every case

$$\forall i, j \in [8] : C_8(i, j) = 1 \Rightarrow |i - j| \geq 3$$

Every contact in the example is at least 3 apart in sequence. (re-checked on all 64 cases.)

T95

compiled check of every case

$$\forall i, j \in [8] : C_8(i, j) = 1 \Rightarrow i - j \leq 5 \vee j - i \leq 5$$

A weak bound, stated as written: for every contact, $i - j \leq 5$ or $j - i \leq 5$. (re-checked on all 64 cases.)

T96

compiled check of every case

$$\forall p, q \in P, p \neq q : d(p, q) \geq 2$$

No two contact cells of the example are one bit apart, so Quine–McCluskey cannot merge any of them: the example is already minimal. (re-checked on all 4,096 cases.)

T97

compiled check of every case

$$d(c_{03}, c_{14}) = 2, d(c_{03}, c_{25}) = 4, d(c_{03}, c_{05}) = 2, d(c_{14}, c_{25}) = 2, d(c_{14}, c_{05}) = 2, d(c_{25}, c_{05}) = 2$$

The six pairwise distances between the example's contact cells: 2, 4, 2, 2, 2, 2. (a closed statement, evaluated once.)

T98

compiled check of every case

$$d(g(k), g(k+1)) = 1 \text{ for } k = 0, 1, \dots, 6$$

Neighbouring residue indices 0–7 are one bit apart in Gray code. (a closed statement, evaluated once.)

A.5 The five-bit cube (8 theorems)

T99

kernel-checked, no axioms

$$\forall a, b \in [20] : \rho(a) = \rho(b) \Rightarrow a = b$$

The raw group-order index 0–19 identifies the amino acid. (re-checked on all 400 cases.)

T100

compiled check of every case

$$N_1 = 80, N_2 = 132, N_3 = 112, N_4 = 48, N_5 = 8$$

Over all 380 ordered pairs of different amino acids, the code distances 1–5 occur 80, 132, 112, 48 and 8 times. (a closed statement, evaluated once.)

T101

compiled check of every case

$$\frac{1}{2}N_1 = 40, \frac{1}{2}N_2 = 66, \frac{1}{2}N_3 = 56, \frac{1}{2}N_4 = 24, \frac{1}{2}N_5 = 4$$

The 190 unordered pairs: 40 at distance 1, 66 at 2, 56 at 3, 24 at 4 and 4 at 5. (a closed statement, evaluated once.)

T102

compiled check of every case

$$\forall n \in [32] : \#\{m \in [32] \mid d(n, m) = 1\} = 5$$

Every corner of the 5-bit cube has exactly five one-bit neighbours. (re-checked on all 32 cases.)

T103

compiled check of every case

$$\frac{1}{2}N_1 \leq 50$$

40 one-bit pairs is within the handshake bound of 50 for any 20 corners of the cube. (a closed statement, evaluated once.)

T104

compiled check of every case

$$\frac{1}{2} \#\{(p, q) \in [32]^2 \mid p \neq q, d(p, q) = 1\} = 80$$

The 5-bit cube has 80 edges. (a closed statement, evaluated once.)

T105

compiled check of every case

$$100 \cdot \frac{1}{2}N_1 / E = 50, \text{ where } E = \frac{1}{2} \#\{(p, q) \in [32]^2 \mid p \neq q, d(p, q) = 1\}$$

The amino-acid code uses exactly half (40 of 80) of the cube's edges. (a closed statement, evaluated once.)

T106

compiled check of every case

$$\forall i \in [19] : d(g(i), g(i+1)) = 1$$

Consecutive raw indices 0–19 are one bit apart after Gray coding. (re-checked on all 19 cases.)

A.6 Contact circuits (12 theorems)**T107**

kernel-checked, standard axioms

$$\forall x \in [31] : d(g(x), g(x+1)) = 1$$

On five bits, consecutive Gray codes 0–31 differ in one bit. (re-checked on all 31 cases.)

T108

kernel-checked, standard axioms

$$\forall a, b \in [32] : d(g(a), g(b)) = d(g(b), g(a))$$

Distance between codes does not depend on the order of the pair. (re-checked on all 1,024 cases.)

T109

compiled check of every case

$$\#\{(a, b) \in [20]^2 \mid d(g(a), g(b)) = 1\} = 80$$

Exactly 80 of the 400 ordered amino-acid pairs are one bit apart: a random pair is one bit apart with probability 0.200. (a closed statement, evaluated once.)

T110

compiled check of every case

$$\left(\#\{(a, b) \in [20]^2 \mid d(g(a), g(b)) = k\} \right)_{k=0}^5 = (20, 80, 132, 112, 48, 8)$$

The whole ordered distance distribution, distances 0–5: 20, 80, 132, 112, 48, 8. (a closed statement, evaluated once.)

T111

kernel-checked, no axioms

$$\sigma(6) = \sigma(10) = 0, \sigma(11) = \sigma(20) = 1, \sigma(21) = \sigma(30) = 2, \sigma(31) = \sigma(266) = 3$$

The four separation bins start at 6, 11, 21 and 31. (a closed statement, evaluated once.)

T112

kernel-checked, standard axioms

$$\forall a, b \in \mathbb{N} : a \leq b \Rightarrow \sigma(a) \leq \sigma(b)$$

Further apart in sequence never means a lower separation bin. (proved for every natural number; the browser re-runs a bounded range.)

T113

kernel-checked, standard axioms

$$\forall g_i, g_j \in [8], s \in [4] : 4(8g_i + g_j) + s < 256$$

A (group, group, separation) cell index is below 256. (re-checked on all 256 cases.)

T114

kernel-checked, standard axioms

$$\begin{aligned} \forall g_i, g_j, g'_i, g'_j \in [8], s, s' \in [4] : \\ 4(8g_i + g_j) + s &= 4(8g'_i + g'_j) + s' \\ \Rightarrow g_i = g'_i \wedge g_j = g'_j \wedge s &= s' \end{aligned}$$

Different (group, group, separation) triples get different cells. (re-checked on all 65,536 cases.)

T115

compiled check of every case

$$\forall c \in [4] : T(c) = 1 \Rightarrow c \in Q_2(0, 1)$$

Completeness, on a worked example: every 1-cell is covered by some cube of the cover. (a closed statement, evaluated once.)

T116

compiled check of every case

$$\forall c \in [4] : T(c) = 0 \Rightarrow c \notin Q_2(0, 1)$$

Soundness, on the same example: no cube of the cover touches a 0-cell. (a closed statement, evaluated once.)

T117

compiled check of every case

$$\forall v, r \in [32], c \in [1024] : c \in Q_{10}(32v + r, 0) \Leftrightarrow c = 32v + r$$

On the padded 32×32 grid, a rule with all ten bits fixed fires on exactly one cell: its own. (re-checked on all 1,048,576 cases.)

T118

compiled check of every case

$$\forall v, r \in [20], c \in [1024] : c \in Q_{10}(32v + r, 0) \Rightarrow \lfloor c/32 \rfloor < 20 \wedge c \bmod 32 < 20$$

Padding is inert for exact rules: a fully fixed rule about real residues (row and column below 20) fires only inside the 20×20 block, never on padding. (re-checked on all 409,600 cases.)

A.7 Sequence circuits (13 theorems)**T119**

compiled check of every case

$$\forall u \in [9], v, i \in [256] \text{ with } 2^u \mid v : i \in Q_8(v, 2^u - 1) \Leftrightarrow v \leq i < v + 2^u$$

With plain-binary positions, a cube whose free bits are the low ones is exactly an unbroken stretch of the chain: positions v to $v + 2^u - 1$. (re-checked on all 589,824 cases.)

T120

compiled check of every case

$$\forall u \in [9], v \in [256] \text{ with } 2^u \mid v, v + 2^u \leq 256 : \#\{i \in [256] \mid i \in Q_8(v, 2^u - 1)\} = 2^u$$

Such a cube covers exactly 2^u positions, so a block's length can be read off its free bits. (re-checked on all 2,304 cases.)

T121

compiled check of every case

$$\begin{aligned} &\forall u, w \in [5], v_i, v_j, i, j \in [16] \\ &\text{with } 2^u \mid v_i, 2^w \mid v_j : \\ &16i + j \in Q_8(16v_i + v_j, \\ &16(2^u - 1) + 2^w - 1) \\ &\Leftrightarrow v_i \leq i < v_i + 2^u \wedge v_j \leq j < v_j + 2^w \end{aligned}$$

A contact-map cube with low free bits in both fields is a block: one chain segment against another. (re-checked on all 1,638,400 cases.)

T122

compiled check of every case

$$\begin{aligned} &\forall u \in [5], v \in [16] \text{ with } 2^u \mid v : \\ &\exists s \in [16], 2^u \mid s : \\ &\{i \in [16] \mid g(i) \in Q_4(v, 2^u - 1)\} \\ &= \{i \in [16] \mid s \leq i < s + 2^u\} \end{aligned}$$

Under Gray-coded positions the same cube is still an unbroken stretch, just a relabelled one. (re-checked on all 80 cases.)

T123

compiled check of every case

$$\exists v \in [16] \text{ with } 4 \mid v : \{i \in [16] \mid g(i) \in Q_4(v, 3)\} \neq \{i \in [16] \mid i \in Q_4(v, 3)\}$$

...but not always the same stretch: binary and Gray disagree about which segment a cube names, so the two must never be mixed. (an existence statement: the check finds a witness.)

T124

compiled check of every case

$$\forall i, i' \in [64], a, a' \in [32] : 32i + a = 32i' + a' \Rightarrow i = i' \wedge a = a'$$

Packing (position, residue) into one number is lossless, so the sequence can be read back from its circuit. (re-checked on all 4,194,304 cases.)

T125

compiled check of every case

$$\forall i, i', a, a' \in [32], i \neq i', a \neq a' : d(32i + a, 32i' + a') \geq 2$$

Two cells with different positions and different residues are never one bit apart, so minimisation can never merge across residues. (re-checked on all 1,048,576 cases.)

T126

compiled check of every case

$$\forall i, a \in [32], k \in [5] : 32i + a \neq 32i + (a \oplus 2^k)$$

Flipping any one residue bit at a fixed position gives a different cell. (re-checked on all 5,120 cases.)

T127

compiled check of every case

$$\forall x, y \in [32] : g(x \oplus y) = g(x) \oplus g(y)$$

The Gray code is linear over GF(2): $g(x \oplus y) = g(x) \oplus g(y)$. (re-checked on all 1,024 cases.)

T128

compiled check of every case

$$\forall x, y \in [32] : g(x) \oplus g(y) = g(x \oplus y)$$

The same identity read the other way: XOR of codes is the code of the XOR. (re-checked on all 1,024 cases.)

T129

compiled check of every case

$$\forall x \in [32] : g(x \oplus 1) = g(x) \oplus 1$$

Flipping the lowest bit commutes with Gray coding, so 2x2 pooling groups the same pairs either way. (re-checked on all 32 cases.)

T130

compiled check of every case

$$\exists x, y \in [32] : d(x, y) \neq d(g(x), g(y))$$

What the Gray code does change: some pair of numbers sits at a different Hamming distance after coding. (an existence statement: the check finds a witness.)

T131

compiled check of every case

$$\forall x, y \in [32] : g(x) = g(y) \Rightarrow x = y$$

On five bits the Gray code is a bijection, so recoding a map only permutes its cells. (re-checked on all 1,024 cases.)